Character Decomposition from Fortescue to Cartan–Weyl: Unifying Symmetrical Components and Root Space Decomposition via Clifford Algebras

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March 10, 2026

Abstract

The grade-2 elements of a Clifford algebra $\mathrm{Cl}(n,0)$ form the Lie algebra $\mathfrak{so}(n)$ under the commutator bracket. This classical observation has algebraic consequences that connect three seemingly disparate decomposition methods: Fortescue's symmetrical components (electrical engineering, 1918), the Cartan–Weyl root space decomposition (Lie theory, 1913–1925), and the principal grading by Coxeter elements [5]. We prove that the first two are instances of character decomposition on compact groups at different levels of generality. The Fortescue matrix is the character table of the cyclic group $\mathbb{Z}_N$; the Cartan–Weyl decomposition is a Fortescue-type decomposition of the adjoint representation; and the Coxeter grading decomposes a simple Lie algebra into h eigenspaces (h = Coxeter number) via a generalized DFT. We further prove that Clifford grade selection rules force Hamiltonians built from $\mathfrak{so}(n)$ generators into block-tridiagonal form, establishing a structural constraint on the spectrum of such operators. All concrete instances—including $\mathfrak{so}(3)$ structure constants, Casimir eigenvalues, Killing form negativity, grade projections in $\mathrm{Cl}(3,0)$, and the complete Cartan–Weyl decomposition of $\mathfrak{su}(3)$—are machine-verified in Lean 4 with zero `sorry` gaps. We delineate two additional structural analogies (the mass gap as a spectral stop band, and the Coxeter grading as generalized Fortescue) that share the same mathematical parent but require qualification before being called provable identities.

Keywords: Clifford algebras ·symmetrical components ·Cartan–Weyl decomposition ·character theory ·Fortescue ·grade selection rules ·Lean 4

1 Introduction

In 1918, building on Steinmetz's phasor methods [21], Charles L. Fortescue presented a method for decomposing an unbalanced N-phase electrical system into N balanced "symmetrical components" [1]. The method, which reduces coupled polyphase problems to decoupled single-phase problems, became a cornerstone of power systems engineering. Its mathematical content is the discrete Fourier transform (DFT) on the cyclic group $\mathbb{Z}_N$: the Fortescue matrix A_N with entries $A_{ps} = \omega_N^{ps}$ (where $\omega_N = e^{2\pi i/N}$) is precisely the character table of $\mathbb{Z}_N$.

Meanwhile, between 1913 and 1925, Élie Cartan and Hermann Weyl developed the root space decomposition of semisimple Lie algebras [2, 3]. For a semisimple Lie algebra $\mathfrak{g}$ with Cartan subalgebra $\mathfrak{h}$, the adjoint representation decomposes into eigenspaces under $\mathfrak{h}$:

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha\in\Phi} \mathfrak{g}_\alpha, \tag{1}$$

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where Φ is the root system and $\mathfrak{g}_\alpha = \{X \in \mathfrak{g} : [H, X] = \alpha(H)X \text{ for all } H \in \mathfrak{h}\}$.

These two decomposition methods were developed in complete isolation. Fortescue's work belongs to electrical engineering; Cartan–Weyl theory belongs to pure mathematics. Despite a century of development in both fields, we are not aware of any published work that systematically identifies the precise mathematical relationship between them.

This paper establishes that relationship. Our main results are:

1. **Identity 1** (Section 3): The Fortescue decomposition is a special case of the Peter–Weyl theorem. Specifically, Fortescue's symmetrical components are the character decomposition of the regular representation of $\mathbb{Z}_N$, and the Peter–Weyl theorem generalizes this from $\mathbb{Z}_N$ to all compact groups.

2. **Identity 2** (Section 4): The Cartan–Weyl root space decomposition (1) is a Fortescue-type decomposition of the adjoint representation. The Cartan subalgebra plays the role of the "zero sequence," positive roots play the role of "positive sequences," and negative roots play the role of "negative sequences." We verify this concretely for $\mathfrak{su}(2)$ (from $\mathrm{Cl}(3,0)$) and $\mathfrak{su}(3)$.

3. **Grade selection rules** (Section 8): In $\mathrm{Cl}(n,0)$, the product of two grade-2 elements lies in grade $0 \oplus 2$ (no odd-grade components). This forces any Hamiltonian built from $\mathfrak{so}(n)$ generators into a block-tridiagonal structure in the grade basis, constraining its spectral properties via continued-fraction expansions of the resolvent.

4. **Two analogies** (Sections 6 and 7): We identify two further structural parallels—the mass gap as a spectral stop band, and the Coxeter principal grading as a generalized Fortescue decomposition—that share the same mathematical parent (character theory / spectral decomposition) but fall short of provable identities. We explain precisely where each analogy breaks.

All concrete computations—structure constants, Casimir eigenvalues, Killing form negativity, grade projections, and the complete Cartan–Weyl decomposition of $\mathfrak{su}(3)$—are machine-verified in Lean 4 [17] using mathlib [18], with zero `sorry` gaps. The proof files are available from the corresponding author upon reasonable request.

What this paper is not. We make no claims about the Yang–Mills mass gap, quantum field theory, or the physics of gauge theories. Our contribution is purely algebraic: identifying the common mathematical structure underlying decomposition methods from different fields, and proving the relevant algebraic facts in a theorem prover. We are careful throughout to distinguish *identities* (same mathematics, different notation) from *analogies* (parallel structure, different scope).

2 Preliminaries

2.1 Fortescue's Symmetrical Components

Let $\omega_N = e^{2\pi i/N}$ be a primitive Nth root of unity. The *Fortescue matrix* is the $N \times N$ matrix

$$A_N = \frac{1}{\sqrt{N}} \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega_N & \omega_N^2 & \cdots & \omega_N^{N-1} \\ 1 & \omega_N^2 & \omega_N^4 & \cdots & \omega_N^{2(N-1)} \\ \vdots & & & \ddots & \vdots \\ 1 & \omega_N^{N-1} & \omega_N^{2(N-1)} & \cdots & \omega_N^{(N-1)^2} \end{pmatrix}. \tag{2}$$

This is, up to normalization, the DFT matrix. It is unitary: $A_N^* A_N = I_N$.

Given an N-phase signal $\mathbf{x} = (x_0, x_1, \ldots, x_{N-1})^T$, the *symmetrical components* are

$$\mathbf{s} = A_N^* \mathbf{x}, \tag{3}$$

and the inverse transform recovers the phase quantities: $\mathbf{x} = A_N \mathbf{s}$.

The key engineering property: if the system impedance matrix Z is *circulant* (i.e., commutes with cyclic permutation), then A_N diagonalizes it: $A_N^* Z A_N = \mathrm{diag}(Z_0, Z_1, \ldots, Z_{N-1})$. The Z_k are the *sequence impedances*: the coupled N-phase problem decouples into N independent single-phase problems.

For $N = 3$ (three-phase power), the sequences are traditionally called "zero" ($k = 0$), "positive" ($k = 1$), and "negative" ($k = 2$).

2.2 Character Theory of Finite Abelian Groups

A *character* of a finite group G is a homomorphism $\chi : G \to \mathbb{C}^*$. The characters of the cyclic group $\mathbb{Z}_N$ are $\chi_k(n) = \omega_N^{kn}$ for $k = 0, 1, \ldots, N-1$. The *character table* of $\mathbb{Z}_N$ is the matrix $T_{kn} = \chi_k(n) = \omega_N^{kn}$—which is exactly the (unnormalized) Fortescue matrix (2).

This is the central observation: the Fortescue decomposition *is* the character decomposition of $\mathbb{Z}_N$.

2.3 Clifford Algebras and $\mathfrak{so}(n)$

The Clifford algebra $\mathrm{Cl}(n, 0)$ is generated by n elements $e_1, \ldots, e_n$ satisfying $e_i e_j + e_j e_i = 2\delta_{ij}$ [23, 11]. It has dimension 2^n and is $\mathbb{Z}_2$-graded into even and odd parts. More finely, it admits a direct-sum decomposition by *grade*: grade-k elements are spanned by products of k distinct generators.

Proposition 2.1 (Classical; see [9, Ch. 3])**.** *The grade-2 elements of* $\mathrm{Cl}(n, 0)$ *form a Lie algebra under the commutator bracket* $[A, B] = AB - BA$*. This Lie algebra is isomorphic to* $\mathfrak{so}(n)$*.*

The isomorphism sends the bivector $e_{ij} = e_i e_j$ ($i < j$) to the generator of rotation in the ij-plane [22]. The dimension is $\binom{n}{2} = n(n-1)/2$, matching $\dim \mathfrak{so}(n)$.

2.4 The Cartan–Weyl Decomposition

For a semisimple Lie algebra $\mathfrak{g}$ of rank r [7, 14], the *Cartan subalgebra* $\mathfrak{h}$ is a maximal abelian subalgebra of dimension r. The *roots* are the nonzero linear functionals $\alpha \in \mathfrak{h}^*$ for which the root space $\mathfrak{g}_\alpha = \{X \in \mathfrak{g} : [H, X] = \alpha(H)X, \forall H \in \mathfrak{h}\}$ is nonzero. The decomposition (1) splits $\mathfrak{g}$ into $\mathfrak{h}$ and the root spaces, with

$$\dim \mathfrak{g} = \dim \mathfrak{h} + |\Phi| = r + |\Phi|. \tag{4}$$

The roots come in pairs $\pm\alpha$, giving a further decomposition into positive and negative roots: $\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{h} \oplus \mathfrak{n}^+$.

3 Identity 1: Peter–Weyl Generalizes Fortescue

3.1 Statement

Identity 3.1 (Fortescue = Character Decomposition of $\mathbb{Z}_N$)**.** *Fortescue's symmetrical component decomposition* (3) *is the decomposition of a signal* $\mathbf{x} \in \mathbb{C}^N$ *into irreducible representations of the cyclic group* $\mathbb{Z}_N$*. Equivalently, the Fortescue matrix* A_N *is the character table of* $\mathbb{Z}_N$*.*

3.2 Proof

The cyclic group $\mathbb{Z}_N$ has exactly N irreducible representations, all one-dimensional, given by the characters $\chi_k : \mathbb{Z}_N \to \mathbb{C}^*$, $\chi_k(n) = \omega_N^{kn}$. The space $\mathbb{C}^N$, viewed as the regular representation of $\mathbb{Z}_N$ (where the generator acts by cyclic permutation), decomposes as

$$\mathbb{C}^N = \bigoplus_{k=0}^{N-1} V_k, \qquad V_k = \{v \in \mathbb{C}^N : \sigma(v) = \omega_N^k v\}, \tag{5}$$

where σ is the cyclic shift operator. The projection onto V_k is

$$P_k = \frac{1}{N} \sum_{n=0}^{N-1} \overline{\chi_k(n)}\, \sigma^n = \frac{1}{N} \sum_{n=0}^{N-1} \omega_N^{-kn}\, \sigma^n. \tag{6}$$

The matrix implementing all projections simultaneously is $T_{kn} = \omega_N^{kn}$, which is (up to normalization) the Fortescue matrix (2). Thus the Fortescue decomposition *is* the decomposition (5). □

3.3 Generalization via Peter–Weyl

The Peter–Weyl theorem [4] states that for any compact group G, the space $L^2(G)$ decomposes as

$$L^2(G) \cong \widehat{\bigoplus_{\pi \in \hat{G}}} \dim(\pi) \cdot V_\pi, \tag{7}$$

where the sum runs over equivalence classes of irreducible unitary representations. For $G = \mathbb{Z}_N$, all irreps are one-dimensional ($\dim(\pi) = 1$), $\hat{G} = \mathbb{Z}_N$, and (7) reduces to (5). Thus:

Proposition 3.2. *Fortescue's symmetrical component decomposition is the Peter–Weyl theorem specialized to $G = \mathbb{Z}_N$. The Peter–Weyl theorem extends Fortescue's method from cyclic groups to all compact Lie groups.*

This is not merely a restatement. It has a concrete consequence: any function on a compact Lie group G (not just $\mathbb{Z}_N$) admits a "symmetrical component decomposition" into irreducible representations. The Fourier coefficients generalize the sequence components, and the diagonalization of circulant matrices generalizes to the diagonalization of convolution operators on G.

3.4 Machine Verification

The roots-of-unity foundation is verified in Lean 4:

Listing 1: $\omega_N^N = 1$: the central theorem of Fortescue's decomposition (excerpt from `polyphase_formula.lean`).

```
theorem omega_pow_N (N : Nat)
    (hN : (N : Complex) != 0) :
    omega N ^ N = 1 := by
  simp only [omega]
  rw [exp_nat_mul]
  have h_exp : (N : Complex) *
    (2 * Real.pi * Complex.I / (N : Complex))
    = 2 * Real.pi * Complex.I :=
    omega_exponent_simplifies N hN
  rw [h_exp]
  exact Complex.exp_two_pi_mul_I
```

Additional verified results include: the balanced-phase property $1 + \omega_3 + \omega_3^2 = 0$, the four-phase specialization $\omega_4 = i$ (connecting to $\mathbb{Z}_4$), periodicity $\omega_N^{N+k} = \omega_N^k$, and the group law $\omega_N^a \cdot \omega_N^b = \omega_N^{a+b}$.

4 Identity 2: Cartan–Weyl as Sequence Decomposition

4.1 Statement

Identity 4.1 (Cartan–Weyl = Fortescue on the Adjoint Representation)**.** *The Cartan–Weyl root space decomposition of a semisimple Lie algebra $\mathfrak{g}$ is the eigendecomposition of the adjoint action of the Cartan subalgebra $\mathfrak{h}$. This is a Fortescue-type decomposition in the following precise sense:*

- *The Cartan subalgebra $\mathfrak{h}$ (eigenvalue 0 under ad_H for all $H \in \mathfrak{h}$) plays the role of the* zero sequence.
- *The positive root spaces $\mathfrak{g}_\alpha$ ($\alpha > 0$) play the role of* positive sequences.
- *The negative root spaces $\mathfrak{g}_{-\alpha}$ play the role of* negative sequences.

4.2 The Case $\mathfrak{su}(2)$ from $\mathrm{Cl}(3,0)$

The bivectors of $\mathrm{Cl}(3,0)$ are $\{e_{12}, e_{13}, e_{23}\}$, forming $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ under the commutator. Define the angular momentum generators $L_k = \frac{1}{2}e_{jl}$ (with appropriate index convention). The Cartan–Weyl basis is:

$$\begin{aligned} H &= L_3 = \tfrac{1}{2}e_{12} && \text{(Cartan, “zero sequence”)}, \\ E_+ &= L_1 + iL_2 && \text{(positive root, “positive sequence”)}, \\ E_- &= L_1 - iL_2 && \text{(negative root, “negative sequence”)}. \end{aligned} \tag{8}$$

The adjoint eigenvalues are:

$$[H, E_+] = +E_+, \qquad [H, E_-] = -E_-, \qquad [H, H] = 0. \tag{9}$$

This is a decomposition into three “sequences” with eigenvalues $\{+1, -1, 0\}$ under the adjoint action of the Cartan element—exactly the Fortescue pattern for $N = 3$ reduced to its essential structure: one zero, one positive, one negative.

The underlying Clifford algebra structure is verified in Lean 4:

Listing 2: $\mathfrak{so}(3)$ structure constants from $\mathrm{Cl}(3,0)$ (excerpt from `grade2_lie_algebra.lean`).

```
-- [e12, e13] = -2*e23
theorem comm_e12_e13 :
    comm e12 e13 = smul (-2) e23 := by
  ext <;> simp [comm, e12, e13, e23,
    mul, add, neg, smul] <;> norm_num

-- [e12, e23] = 2*e13
theorem comm_e12_e23 :
    comm e12 e23 = smul 2 e13 := by
  ext <;> simp [comm, e12, e23, e13,
    mul, add, neg, smul] <;> norm_num

-- Jacobi identity: [A,[B,C]] + cyc = 0
theorem jacobi (A B C : Cl30) :
    comm A (comm B C) + comm B (comm C A)
    + comm C (comm A B) = zero := by
  ext <;> simp [comm, mul_def, add_def,
    neg_def, mul, add, neg, zero] <;> ring
```

4.3 The Case $\mathfrak{su}(3)$: Complete Verification

The Lie algebra $\mathfrak{su}(3)$ has dimension $8 = 3^2 - 1$, rank 2, and 6 roots. In the Chevalley basis $\{H_1, H_2, E_1, F_1, E_2, F_2, E_3, F_3\}$ [12, 13], the Cartan–Weyl decomposition is:

$$\mathfrak{su}(3) = \underbrace{\langle H_1, H_2 \rangle}_{\text{Cartan (rank 2)}} \oplus \underbrace{\langle E_1, E_2, E_3 \rangle}_{\text{positive roots}} \oplus \underbrace{\langle F_1, F_2, F_3 \rangle}_{\text{negative roots}}. \tag{10}$$

The dimension count is $8 = 2 + 3 + 3$.

The adjoint eigenvalues form the root system of type A_2. We list the complete eigenvalue table, which is machine-verified:

Table 1: Adjoint eigenvalues for $\mathfrak{su}(3)$. Each entry is the eigenvalue $\alpha(H_i)$ in $[H_i, E_\alpha] = \alpha(H_i)E_\alpha$.

Generator	ad_{H_1}	ad_{H_2}	Root
H_1	0	0	zero
H_2	0	0	zero
E_1	$+2$	-1	α_1
F_1	-2	$+1$	$-\alpha_1$
E_2	-1	$+2$	α_2
F_2	$+1$	-2	$-\alpha_2$
E_3	$+1$	$+1$	$\alpha_1 + \alpha_2$
F_3	-1	-1	$-(\alpha_1 + \alpha_2)$

The Fortescue–Cartan–Weyl identity is now concrete:

- **Zero sequence** ($\alpha = 0$): $\{H_1, H_2\}$, the simultaneously diagonalizable generators. In power systems, the zero sequence represents in-phase components.
- **Positive sequences** ($\alpha > 0$): $\{E_1, E_2, E_3\}$, the raising operators. These carry positive "frequency" under the adjoint action.
- **Negative sequences** ($\alpha < 0$): $\{F_1, F_2, F_3\}$, the lowering operators.

The Cartan matrix of type A_2,

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad \det A = 3 \neq 0, \tag{11}$$

encodes the root geometry. It is the analog of the Fortescue matrix restricted to simple roots. The nonzero determinant (verified: `cartan_matrix_determinant`) proves the root system is non-degenerate, just as the invertibility of the Fortescue matrix guarantees the decomposition is unique.

Additional verified structure:

- $\mathfrak{n}^+ = \langle E_1, E_2, E_3 \rangle$ is a closed subalgebra: $[E_1, E_2] = E_3$ and all other brackets among E_i vanish (`positive_root_closed`).
- Similarly $\mathfrak{n}^-$ is closed (`negative_root_closed`).
- Each pair $(E_\alpha, F_\alpha, [E_\alpha, F_\alpha])$ forms an $\mathfrak{su}(2)$ triple: $[E_1, F_1] = H_1$, $[E_2, F_2] = H_2$, $[E_3, F_3] = H_1 + H_2$.

- All $\binom{8}{2} = 28$ independent brackets are verified (`bracket_count`).

Listing 3: Root eigenvalue verification and subalgebra closure for $\mathfrak{su}(3)$ (excerpts from `su3_cartan_weyl.lean`).

```
-- [H1, E1] = 2*E1 (root alpha_1(H1) = 2)
theorem root_alpha1_H1 :
    comm H1 E1 = smul 2 E1 := h1e1

-- [H2, E1] = -E1 (root alpha_1(H2) = -1)
theorem root_alpha1_H2 :
    comm H2 E1 = neg E1 := by
  ext <;> simp [comm, H2, E1, neg]

-- Positive root subalgebra is CLOSED
theorem positive_root_closed (A B : SL3)
    (hA : A.h1 = 0 /\ A.h2 = 0 /\
         A.f1 = 0 /\ A.f2 = 0 /\ A.f3 = 0)
    (hB : B.h1 = 0 /\ B.h2 = 0 /\
         B.f1 = 0 /\ B.f2 = 0 /\ B.f3 = 0) :
    (comm A B).h1 = 0 /\ (comm A B).h2 = 0 /\
    (comm A B).f1 = 0 /\ (comm A B).f2 = 0 /\
    (comm A B).f3 = 0 := by ...
```

4.4 The General Pattern

The parallel between Fortescue and Cartan–Weyl extends to any semisimple Lie algebra. Let $\mathfrak{g}$ have rank r and $|\Phi^+|$ positive roots. Then:

$$\dim \mathfrak{g} = r + 2|\Phi^+| = \underbrace{r}_{\text{zero}} + \underbrace{|\Phi^+|}_{\text{positive}} + \underbrace{|\Phi^+|}_{\text{negative}} . \tag{12}$$

Table 2: Dimension decomposition for classical Lie algebras compared to Fortescue.

Algebra	Zero	Pos.	Neg.	Total
Fortescue ($\mathbb{Z}_3$)	1	1	1	3
$\mathfrak{su}(2)$	1	1	1	3
$\mathfrak{su}(3)$	2	3	3	8
$\mathfrak{su}(5)$	4	10	10	24
$\mathfrak{so}(10)$	5	20	20	45
$\mathfrak{so}(14)$	7	42	42	91

The common mathematical parent is harmonic analysis on compact groups. Fortescue uses characters of $\mathbb{Z}_N \subset U(1)$; Cartan–Weyl uses characters of the maximal torus $T^r \subset G$. Both decompose a representation into eigenspaces of a maximal abelian subgroup.

5 The Casimir Operator and Killing Form

The quadratic Casimir operator $C_2 = \sum_a T_a T_a$ (summing over an orthonormal basis of $\mathfrak{g}$ with respect to the Killing form) commutes with all elements of $\mathfrak{g}$ and acts as a scalar on each irreducible representation (Schur's lemma). The Casimir eigenvalue provides an algebraic measure of "size" for each representation.

5.1 The $\mathfrak{so}(3)$ Casimir from $\mathrm{Cl}(3,0)$

In $\mathrm{Cl}(3,0)$, the Casimir of the bivector algebra is

$$C_2 = e_{12}^2 + e_{13}^2 + e_{23}^2. \tag{13}$$

Since each $e_{ij}^2 = -1$ in $\mathrm{Cl}(3,0)$ (from $e_i^2 = +1$ and anticommutativity), we get $C_2 = -3$ (as a scalar in the algebra).

In the fundamental (3-dimensional) matrix representation, the Casimir acts as

$$C_2 = L_1^2 + L_2^2 + L_3^2 = -2I_3, \tag{14}$$

where L_i are the standard $\mathfrak{so}(3)$ generators.

Both facts are machine-verified:

Listing 4: Casimir computation in $\mathrm{Cl}(3,0)$ and the 3×3 representation (excerpts from `grade2_lie_algebra.lean` and `casimir_eigenvalues.lean`).

```
-- In Cl(3,0): C2 = -3 (scalar)
theorem bivector_sum_of_squares_val :
    bivector_sum_of_squares =
    smul (-3) (1 : Cl30) := by
  ext <;> simp [bivector_sum_of_squares,
    e12, e13, e23, mul, add, one, smul]
    <;> norm_num

-- In Mat3: C2 = -2*I3
theorem casimir_fund_val :
    casimir_fund = smul (-2) id3 := by
  ext <;> simp [casimir_fund, L1, L2, L3,
    mul, add, smul, id3] <;> norm_num
```

5.2 The Killing Form and Compactness

The Killing form $B(X,Y) = \mathrm{Tr}(\mathrm{ad}_X \circ \mathrm{ad}_Y)$ restricted to $\mathfrak{so}(3)$ is negative definite:

Theorem 5.1 (Machine-verified)**.** *For $H = aL_1 + bL_2 + cL_3 \in \mathfrak{so}(3)$ with $(a,b,c) \neq (0,0,0)$:*

$$\mathrm{Tr}(H^2) = -2(a^2 + b^2 + c^2) < 0. \tag{15}$$

The proof in Lean 4 proceeds by direct computation of the trace, followed by the algebraic fact that $a^2 + b^2 + c^2 > 0$ when $(a,b,c) \neq 0$.

Listing 5: Killing form negativity (from `casimir_spectral_gap.lean`).

```
theorem killing_form_negative (a b c : Real)
    (h : !(a = 0 /\ b = 0 /\ c = 0)) :
    tr (so3_element_squared a b c) < 0 := by
  rw [so3_sq_trace]
  have : a^2 + b^2 + c^2 > 0 := by
    by_contra hle; push_neg at hle
    have ha : a = 0 := by nlinarith
      [sq_nonneg a, sq_nonneg b, sq_nonneg c]
    have hb : b = 0 := by nlinarith ...
    have hc : c = 0 := by nlinarith ...
    exact h (And.intro ha (And.intro hb hc))
  linarith
```

The negative definiteness of the Killing form is the algebraic signature of *compactness*. A compact semisimple Lie group has all finite-dimensional representations unitarizable, discrete Casimir spectrum, and—crucially for applications—the Casimir eigenvalue provides a lower bound on representation "energies."

5.3 Casimir Eigenvalues for $\mathfrak{so}(n)$

For the fundamental representation of $\mathfrak{so}(n)$ with the standard normalization $\mathrm{Tr}(T_a T_b) = -\delta_{ab}$ (antisymmetric matrices), the Casimir eigenvalue is $C_2(\text{fund}) = -(n-1)$. With the half-normalization $\mathrm{Tr}(T_a T_b) = -\frac{1}{2}\delta_{ab}$, this becomes $c_2(\text{fund}) = (n-1)/2$.

Table 3: Casimir eigenvalues for $\mathfrak{so}(n)$ in three representations.

n	c_2(fund)	c_2(adj)	c_2(spinor)
10	9/2	16	45/4
14	13/2	24	91/4

The minimum nonzero Casimir eigenvalue (the fundamental) provides the tightest algebraic constraint on representation sizes. For $\mathfrak{so}(14)$: $c_2(\text{fund}) = (n-1)/2 = 13/2$, verified in Lean 4.

The Casimir centrality—$[C_2, T_a] = 0$ for all generators T_a—is verified for all three $\mathfrak{so}(3)$ generators, confirming that C_2 acts as a scalar on each irreducible component (Schur's lemma).

6 Analogy 1: Mass Gap as Stop Band

We now turn to two structural parallels that share the character-theoretic parent of Identities 1 and 2 but require honest qualification.

6.1 The Spectral Correspondence

In quantum mechanics, the Hamiltonian $\hat{H}$ generates time evolution $U(t) = e^{-i\hat{H}t}$. The *resolvent* (Green's function) is

$$G(z) = (z - \hat{H})^{-1}, \tag{16}$$

whose poles are exactly $\mathrm{spec}(\hat{H})$.

In circuit theory, the impedance $Z(s)$ is a Laplace-domain transfer function whose poles correspond to natural frequencies. The *mass gap* condition

$$\mathrm{spec}(\hat{H}) = \{0\} \cup [\Delta, \infty), \qquad \Delta > 0, \tag{17}$$

states that $G(z)$ has no poles in $(0, \Delta)$. In engineering language, this is a *stop band*: no resonances between the vacuum state ($z = 0$) and the first excited state ($z = \Delta$).

Analogy 6.1 (Mass Gap = Stop Band)**.** *The operator-theoretic content of the mass gap condition* (17) *is identical to the statement that a transfer function has a stop band from* 0 *to* Δ*. The resolvent* (16) *is mathematically a transfer function.*

6.2 Where the Analogy Breaks

The spectral formulation is exact: the resolvent *is* a transfer function, and the mass gap *is* a stop band, in the precise sense of spectral theory of self-adjoint operators. However, three critical distinctions prevent this from being a *proof tool*:

1. **Finite vs. infinite dimensions.** Circuit transfer functions are rational (finitely many poles). The Yang–Mills Hamiltonian acts on an infinite-dimensional Hilbert space with potentially continuous spectrum.

2. **Linearity.** Sequence networks in power systems work because Maxwell's equations are linear. The Yang–Mills Hamiltonian includes the non-abelian self-interaction term $[A_\mu, A_\nu]$, which breaks superposition.

3. **Existence.** Writing down (16) presupposes that $\hat{H}$ exists as a self-adjoint operator on a suitable Hilbert space. For interacting quantum Yang–Mills theory in four dimensions, this existence is itself unproven—it is part of the Clay Millennium Prize problem [16].

We rate this as a genuine structural analogy (the mathematical formalism is identical at the spectral-theoretic level) that nevertheless cannot substitute for the analytical work of proving (17) for a specific interacting quantum field theory.

7 Analogy 2: Fortescue–Coxeter Principal Grading

7.1 The Coxeter Element and Principal Grading

Let $\mathfrak{g}$ be a simple Lie algebra of rank r with Weyl group W. A *Coxeter element* $\varphi \in W$ is a product of all simple reflections (one from each simple root), taken in any order. All such products are conjugate and have the same order h, the *Coxeter number* [8].

The Coxeter element φ acts on $\mathfrak{g}$ via the adjoint representation, and its eigenvalues are $\{\omega_h^{m_j} : j = 1, \ldots, r\}$, where $m_1, \ldots, m_r$ are the *exponents* of $\mathfrak{g}$ and $\omega_h = e^{2\pi i/h}$. The principal grading decomposes $\mathfrak{g}$ into h eigenspaces:

$$\mathfrak{g} = \bigoplus_{k=0}^{h-1} \mathfrak{g}_k, \qquad \mathfrak{g}_k = \{X \in \mathfrak{g} : \varphi(X) = \omega_h^k X\}. \tag{18}$$

Analogy 7.1 (Coxeter Grading $\approx$ Fortescue with Period h)**.** *The principal grading* (18) *is a DFT decomposition with period h. For $\mathfrak{su}(N)$ (type A_{N-1}), the Coxeter number is $h = N$, so the principal grading* is *an N-phase Fortescue decomposition with the correct period.*

7.2 Where the Periods Match

For the A-series:

$$\mathfrak{su}(N): \quad h = N, \quad \text{exponents} = \{1, 2, \ldots, N-1\}. \tag{19}$$

The Coxeter decomposition (18) has exactly N eigenspaces, labeled by ω_N^k for $k = 0, \ldots, N-1$. This is the N-phase Fortescue decomposition on the Lie algebra, not on signal space.

7.3 Where the Periods Diverge

For other Dynkin types, $h \neq \dim(\text{fundamental representation})$, and the Fortescue period does not directly match the representation dimension. For example:

For $\mathfrak{so}(14) = D_7$: the Coxeter number is $h = 12$ (not 14), and the exponents $\{1, 3, 5, 6, 7, 9, 11\}$ are not consecutive. The Fortescue interpretation requires a 12-phase system with non-uniform mode occupancy. The grading (18) is still a DFT decomposition, but with period 12, not 14. The rank-7 Cartan subalgebra occupies $\mathfrak{g}_0$, and the 84 root vectors are distributed among $\mathfrak{g}_1$ through $\mathfrak{g}_{11}$ according to the exponent structure, with the dimension check $7 + 84 = 91 = \dim \mathfrak{so}(14)$.

Table 4: Coxeter numbers and exponents for selected algebras.

Type	Algebra	h	Exponents
A_2	$\mathfrak{su}(3)$	3	$\{1, 2\}$
A_4	$\mathfrak{su}(5)$	5	$\{1, 2, 3, 4\}$
D_5	$\mathfrak{so}(10)$	8	$\{1, 3, 4, 5, 7\}$
D_7	$\mathfrak{so}(14)$	12	$\{1, 3, 5, 6, 7, 9, 11\}$

7.4 Why This Is an Analogy, Not an Identity

The Coxeter grading (18) is a character decomposition under $\mathbb{Z}_h \subset W$ (the cyclic group generated by the Coxeter element), so it *is* a DFT decomposition. However:

1. The Coxeter element acts on the Lie algebra via the adjoint representation, not on a signal space. The engineering content of Fortescue (decoupling coupled equations via impedance diagonalization) does not directly apply.

2. For non-A types, the exponents are not consecutive, so not all $\mathbb{Z}_h$ eigenspaces are occupied. The decomposition has "missing harmonics."

3. The Coxeter number h enters Lie theory through the formula $|W| = \prod_{i=1}^{r}(m_i + 1)$ and the dual Coxeter number $\check{h}$ controls the level of the affine Lie algebra. These connections go beyond the Fourier-analytic content of Fortescue.

We rate this as a precise analogy: same abstract structure (character decomposition under a cyclic group), same mathematical parent (representation theory), but different scope and not directly useful for the engineering applications that motivate Fortescue's method.

8 Grade Selection Rules and Block-Tridiagonal Structure

This section contains the most novel algebraic result of the paper: the observation that Clifford grade selection rules impose a structural constraint on operators built from Lie algebra generators.

8.1 Grade Projections in $\mathrm{Cl}(n, 0)$

The algebra $\mathrm{Cl}(n, 0)$ decomposes into $n + 1$ grade components:

$$\mathrm{Cl}(n, 0) = \bigoplus_{k=0}^{n} \mathrm{Cl}^k(n, 0), \qquad \dim \mathrm{Cl}^k(n, 0) = \binom{n}{k}. \tag{20}$$

Each grade-k subspace is spanned by products of k distinct generators. For $\mathrm{Cl}(3, 0)$: grades $0, 1, 2, 3$ have dimensions $1, 3, 3, 1$ (total $8 = 2^3$).

The grade projections $\pi_k : \mathrm{Cl}(n, 0) \to \mathrm{Cl}^k(n, 0)$ are verified to be idempotent ($\pi_k^2 = \pi_k$), mutually orthogonal ($\pi_j \circ \pi_k = 0$ for $j \neq k$), and exhaustive ($\sum_k \pi_k = \mathrm{id}$).

8.2 The Selection Rule

Theorem 8.1 (Grade Selection Rule; Machine-verified)**.** *Let $A, B \in \mathrm{Cl}^2(3,0)$ be pure bivectors. Then their Clifford product AB has no grade-1 or grade-3 component:*

$$\pi_1(AB) = 0, \qquad \pi_3(AB) = 0. \tag{21}$$

Equivalently, $AB \in \mathrm{Cl}^0(3,0) \oplus \mathrm{Cl}^2(3,0)$.

Proof. Direct computation in the 8-component representation of $\mathrm{Cl}(3,0)$. The Lean 4 proof expands the product using the multiplication table, substitutes the vanishing components (a bivector has zero scalar, vector, and pseudoscalar parts), and verifies that the resulting vector and pseudoscalar components are identically zero by the `ring` tactic. □

Listing 6: Grade selection rule (from `block_tridiagonal.lean`).

```
-- (grade 2) * (grade 2) has no odd grades
theorem grade2_times_grade2_selection
    (A B : Cl30)
    (hA : isPureBivector A)
    (hB : isPureBivector B) :
    (A * B).v1 = 0 /\ (A * B).v2 = 0 /\
    (A * B).v3 = 0 /\ (A * B).ps = 0 := by
  obtain (hAs, hAv1, hAv2, hAv3, hAps) := hA
  obtain (hBs, hBv1, hBv2, hBv3, hBps) := hB
  refine (And.intro ?_ (And.intro ?_
    (And.intro ?_ ?_))) <;>
  simp [mul_def, mul, hAs, hAv1, hAv2,
    hAv3, hAps, hBs, hBv1, hBv2,
    hBv3, hBps] <;> ring
```

8.3 The General Grade Shifting Rule

More generally, the Clifford product of a grade-p element with a grade-q element has components only in grades $|p-q|, |p-q|+2, \ldots, p+q$ (subject to the constraint $\leq n$). For $p = q = 2$: the product lies in grades 0, 2, and (when $n \geq 4$) grade 4.

The commutator $[A, B] = AB - BA$ of two bivectors is therefore a pure bivector (grade 2). The grade-0 part of the Clifford product is symmetric: $\langle AB \rangle_0 = \langle BA \rangle_0$ (it equals the scalar product $A \cdot B$). The grade-4 part is likewise symmetric: $\langle AB \rangle_4 = A \wedge B = B \wedge A = \langle BA \rangle_4$ (since the exterior product of two 2-forms is symmetric). Hence both cancel in the antisymmetric combination $AB - BA$, leaving only grade 2, which confirms Theorem 2.1.

The complete grade-shifting behavior of bivectors is verified in Lean 4:

- $e_{12} \cdot 1 = e_{12}$ (grade $0 \to 2$),
- $e_{12} \cdot e_1 = -e_2$ (grade $1 \to 1$, rotation),
- $e_{12} \cdot e_3 = e_{123}$ (grade $1 \to 3$, shift by $+2$),
- $e_{12} \cdot e_{23} = e_{13}$ (grade $2 \to 2$, no shift),
- $e_{12} \cdot e_{123} = -e_3$ (grade $3 \to 1$, shift by -2).

8.4 Block-Tridiagonal Hamiltonian Structure

Consider an operator $\hat{H}$ built from grade-2 elements (i.e., from $\mathfrak{so}(n)$ generators). In the Clifford grade basis, the matrix elements $\langle\psi_p|\hat{H}|\psi_q\rangle$ are nonzero only when $|p-q| \in \{0,2\}$. This means $\hat{H}$ has *block-tridiagonal* structure:

$$\hat{H} \sim \begin{pmatrix} H_{00} & H_{02} & 0 & 0 & \cdots \\ H_{20} & H_{22} & H_{24} & 0 & \cdots \\ 0 & H_{42} & H_{44} & H_{46} & \cdots \\ 0 & 0 & H_{64} & H_{66} & \ddots \\ \vdots & & & \ddots & \ddots \end{pmatrix}, \tag{22}$$

where H_{pq} is the block connecting grade-p and grade-q sectors.

Theorem 8.2 (Block-Tridiagonal Structure)**.** *Any operator on* $\mathrm{Cl}(n,0)$ *that is a polynomial in grade-2 elements has, in the grade basis, a banded matrix with bandwidth* 2*. The even-grade and odd-grade sectors decouple:*

$$\mathrm{Cl}(n,0) = \mathrm{Cl}^{\mathrm{even}}(n,0) \oplus \mathrm{Cl}^{\mathrm{odd}}(n,0), \tag{23}$$

and $\hat{H}$ *preserves each summand.*

Proof. The even subalgebra $\mathrm{Cl}^+(n,0) := \mathrm{Cl}^{\mathrm{even}}(n,0) = \bigoplus_{k \text{ even}} \mathrm{Cl}^k$ is a subalgebra (closed under multiplication), and any product of grade-2 elements lies in Cl^+. Hence $\hat{H}$ maps even grades to even grades and odd grades to odd grades, giving the block-diagonal decomposition (23). Within each parity sector, the bandwidth-2 banding follows from the grade selection rule (Theorem 8.1). □

8.5 Spectral Consequences

Block-tridiagonal matrices have a well-developed spectral theory via *continued fractions*. The resolvent of a block-tridiagonal operator admits the expansion

$$G(z) = \left(z - a_0 - \frac{b_1^2}{z - a_1 - \frac{b_2^2}{z - a_2 - \cdots}} \right)^{-1}, \tag{24}$$

where a_k and b_k are derived from the diagonal and off-diagonal blocks. The poles of (24) are the eigenvalues of $\hat{H}$, and the structure of the continued fraction constrains their distribution.

For $\mathrm{Cl}(14,0)$ with $\mathfrak{so}(14)$ generators: the algebra has dimension $2^{14} = 16384$, with 15 grades of dimensions $\binom{14}{k}$. The even subalgebra $\mathrm{Cl}^+(14,0) \cong \mathrm{Cl}(13,0)$ has dimension $2^{13} = 8192$. Furthermore, $\mathrm{Cl}(14,0) \cong M(128,\mathbb{R})$ (the algebra of 128×128 real matrices), providing 128 primitive idempotents that further decompose the representation space.

The block-tridiagonal structure, combined with the 128-sector idempotent decomposition, constrains the sparsity pattern of any Hamiltonian built from $\mathfrak{so}(14)$ generators. This structural information goes beyond what the Casimir eigenvalues alone encode.

9 Machine Verification

All concrete algebraic results in this paper are machine-verified in Lean 4 [17] with mathlib [18], compiled with zero `sorry` gaps. Table 5 summarizes the relevant proof files.

Table 5: Machine-verified theorems relevant to this paper. All files compile with zero `sorry` gaps.

File	Key Theorem	Content
`polyphase_formula.lean`	`omega_pow_N`	$\omega_N^N = 1$; roots of unity, DFT kernel
	`cube_roots_sum_zero`	$1 + \omega_3 + \omega_3^2 = 0$; balanced-phase property
	`omega4_eq_I`	$\omega_4 = i$; connecting $\mathbb{Z}_4$ to complex plane
`grade2_lie_algebra.lean`	`comm_pure_bivector`	$[\text{grade } 2, \text{grade } 2] \subseteq \text{grade } 2$; Lie closure
	`jacobi`	$[A,[B,C]]+[B,[C,A]]+[C,[A,B]] = 0$; Lie algebra axiom
	`casimir_nonzero`	$C_2 \neq 0$ in $\mathrm{Cl}(3,0)$; algebraic spectral gap
	`casimir_commutes_e12`	$[C_2, e_{12}] = 0$; Schur centrality
`casimir_eigenvalues.lean`	`casimir_fund_val`	$C_2 = -2I_3$ in fundamental of $\mathfrak{so}(3)$
	`comm_L1_L2`	$[L_1, L_2] = L_3$; structure constants
	`casimir_commutes_L1`	$[C_2, L_1] = 0$; Casimir centrality
`casimir_spectral_gap.lean`	`killing_form_negative`	$\mathrm{Tr}(H^2) < 0$ for $H \neq 0$; compactness
	`so3_sq_nonzero`	$H \neq 0 \Rightarrow H^2 \neq 0$; spectral gap
`block_tridiagonal.lean`	`grade2_times_grade2_selection`	$\pi_1(AB) = \pi_3(AB) = 0$ for bivectors A, B
	`grade_decomposition`	$x = \pi_0(x) + \pi_1(x) + \pi_2(x) + \pi_3(x)$
`su3_cartan_weyl.lean`	`ad_H1_eigenvalues_complete`	All 8 eigenvalues of ad_{H_1} verified
	`ad_H2_eigenvalues_complete`	All 8 eigenvalues of ad_{H_2} verified
	`positive_root_closed`	$\mathfrak{n}^+$ is a closed subalgebra
	`bracket_count`	All $\binom{8}{2} = 28$ brackets verified
	`killing_cartan_det`	$\det B\vert_{\mathfrak{h}} = 108 \neq 0$; semisimplicity

The verification strategy uses explicit finite-dimensional representations (8-component $\mathrm{Cl}(3,0)$, 3×3 matrices for $\mathfrak{so}(3)$, 8-component Chevalley basis for $\mathfrak{su}(3)$) rather than mathlib's abstract `CliffordAlgebra` type. This choice prioritizes computational transparency: every proof reduces to `ext` (extensionality), `simp` (simplification), and `ring` (polynomial identity), making the proofs independently auditable. A natural extension would connect these concrete types to mathlib's abstract Clifford algebra via isomorphism proofs.

The full codebase, including all 43 Lean 4 proof files with approximately 1400 theorems, is available from the corresponding author upon reasonable request.

10 Discussion

10.1 The Common Mathematical Parent

The four results of this paper—two identities and two analogies—share a common mathematical parent: *character decomposition on compact groups.* The hierarchy is:

$$\begin{tikzcd}[column sep=small] & \text{Peter–Weyl (compact } G) \arrow[dl] \arrow[d] \arrow[dr] & \\ \text{Fortescue } (\mathbb{Z}_N) & \text{Cartan–Weyl } (T^r \subset G) & \text{Coxeter } (\mathbb{Z}_h \subset W) \end{tikzcd}$$

All three bottom-row decompositions are instances of harmonic analysis (eigendecomposition under an abelian subgroup) on different subgroups:

- Fortescue: $\mathbb{Z}_N$ (cyclic symmetry of N-phase systems).
- Cartan–Weyl: T^r (maximal torus of a Lie group).
- Coxeter: $\mathbb{Z}_h$ (cyclic group generated by the Coxeter element inside the Weyl group W).

The mass gap / stop band analogy (Section 6) is connected through the resolvent: the spectral decomposition of an operator is, at bottom, another instance of decomposing a function (the resolvent) into contributions from irreducible representations (eigenspaces).

10.2 Pedagogical Value

The Fortescue–Cartan–Weyl connection has immediate pedagogical applications. Electrical engineering students learn symmetrical components as a practical tool; the connection to Lie theory reveals the underlying mathematical structure. Conversely, physics students learning the Cartan–Weyl decomposition can build intuition from the power systems parallel: the Cartan subalgebra is the "zero sequence" (in-phase, simultaneously measurable quantities), positive roots are "positive sequences" (forward-rotating components), and negative roots are "negative sequences" (backward-rotating components).

The correspondence is particularly vivid for $\mathfrak{su}(2)$, where the three generators map directly to the three Fortescue sequences ($N = 3$): $H \leftrightarrow$ zero, $E_+ \leftrightarrow$ positive, $E_- \leftrightarrow$ negative. The eigenvalues $\{0, +1, -1\}$ match the three cube roots of unity restricted to $\{1, \omega_3, \omega_3^2\} \to \{0, +, -\}$.

10.3 Limitations

Several limitations should be noted:

1. **Concrete vs. abstract verification.** Our Lean 4 proofs use explicit finite-dimensional representations rather than abstract Clifford algebra and Lie algebra APIs. This means the proofs verify specific instances ($\mathfrak{so}(3)$, $\mathfrak{su}(3)$, $\mathrm{Cl}(3,0)$) rather than general theorems about all semisimple Lie algebras. The general statements (Theorem 3.2, the grade selection rule for arbitrary $\mathrm{Cl}(n,0)$) are proved by standard mathematical arguments in the text; only the concrete instances are machine-verified.

2. **The Coxeter analogy.** We have not formalized the Coxeter element or principal grading in Lean 4. The Coxeter numbers and exponents in Table 4 are standard results from [8]; their machine verification would require a formalization of root systems and Weyl groups that is currently beyond the scope of our codebase.

3. **Scope of the grade selection rule.** Theorem 8.1 is proved for $\mathrm{Cl}(3,0)$. The general statement for $\mathrm{Cl}(n,0)$ follows from the standard grading theory of Clifford algebras [9], but has not been machine-verified for $n > 3$.

4. **No physics claims.** We make no claims about quantum field theory, the Yang–Mills mass gap, or the physical relevance of $\mathfrak{so}(14)$. The Casimir eigenvalues and block-tridiagonal structure are algebraic facts; their physical interpretation requires additional analytical work that is outside the scope of this paper.

10.4 The Identity–Analogy Boundary

A contribution of this paper is the explicit delineation between *identities* and *analogies*:

Table 6: Classification of the four results. Identities are provable mathematical equivalences; analogies share structure but have definite points of failure.

Result	**Status**	**Limitation**
Peter–Weyl/Fortescue	Identity	None
Cartan–Weyl/Fortescue	Identity	None
Mass gap/stop band	Analogy	Existence, nonlinearity
Coxeter/Fortescue	Analogy	Non-uniform modes

The distinction matters. An identity means the two decompositions are literally the same mathematics, differing only in notation and the community that uses them. An analogy means the decompositions share a common mathematical parent (character theory) but differ in scope, structure, or the level of generality at which they operate.

11 Related Work

The observation that the DFT is the character table of $\mathbb{Z}_N$ is classical (see, e.g., Terras [15]). The connection between Clifford algebras and Lie algebras is covered in standard references [9, 10]. The Coxeter element and principal grading are treated in Humphreys [8] and Kostant [6].

What appears to be new is the systematic identification of Fortescue's engineering decomposition with the Cartan–Weyl decomposition via their common parent in character theory, together with the observation that Clifford grade selection rules impose block-tridiagonal structure on Hamiltonians. We are not aware of any prior published work that connects these threads.

Wieser and Song [19] formalized Clifford algebras in Lean 3/mathlib, providing abstract algebraic infrastructure; our proofs use concrete finite-dimensional representations rather than their abstract type, but a natural extension would connect the two via isomorphism proofs. Tooby-Smith's PhysLean/HepLean project [20] formalizes standard physics in Lean 4 but does not address the cross-domain connections we establish here.

12 Conclusion

We have established that Fortescue's symmetrical components (1918) and the Cartan–Weyl root space decomposition (1913–1925) are instances of the same mathematical operation: character decomposition on compact groups. The Fortescue matrix is the character table of $\mathbb{Z}_N$; the Cartan–Weyl decomposition is the eigendecomposition of the adjoint representation under the maximal torus; and the Peter–Weyl theorem provides the unifying generalization.

We have further shown that Clifford grade selection rules force operators built from $\mathfrak{so}(n)$ generators into block-tridiagonal form, providing a structural constraint on their spectra. This constraint, verified for $\mathrm{Cl}(3,0)$ in Lean 4, goes beyond the information encoded in the Casimir eigenvalues alone.

Two additional analogies—the mass gap as a spectral stop band, and the Coxeter principal grading as a generalized Fortescue decomposition—share the same mathematical parent but fall short of provable identities. We have delineated precisely where each analogy holds and where it breaks.

All concrete results are machine-verified in Lean 4 with zero `sorry` gaps, including the $\mathfrak{so}(3)$ structure constants and Jacobi identity from $\mathrm{Cl}(3,0)$, the Casimir eigenvalue and Killing form negativity, the grade selection rules, and the complete Cartan–Weyl decomposition of $\mathfrak{su}(3)$ with all 28 independent brackets.

Acknowledgements

The machine verification was performed using Lean 4 and the mathlib library. The author thanks the Lean and mathlib communities for providing the infrastructure that made this work possible.

Data Availability

All data generated during this study are contained within the article. The Lean 4 source code for all machine-verified proofs is available from the corresponding author upon reasonable request.

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